DELHI PUBLIC SCHOOL NEWTOWN
SESSION: 2022-23
PRE-BOARD EXAMINATION

CLASS: X | **FULL MARKS: 80**
SUBJECT: CHEMISTRY [SET A] | **TIME: 2 HOURS**

General Instructions:

- **The paper consists of <u>six</u> printed pages.**
- **Section A is compulsory. Attempt *<u>any four</u>* questions from Section B.**
- **Answers should be to the point.**
- **Question numbers should be copied carefully while answering the questions.**

SECTION A

(Attempt all questions from this section)

Question 1

Choose one correct answer to the questions from the given options: **[15]**

(i) A weak electrolyte is:

A. Acetic acid
B. Carbon tetrachloride
C. Molten sodium chloride
D. Sugar syrup

(ii) Ionisation potential is maximum for the element is:

A. Noble gases
B. Alkaline earth metals
C. Halogens
D. Alkali metal

(iii) The main component of brass is:

A. Nickel
B. Copper
C. Chromium
D. Aluminium

(iv) The polar covalent compound is:

A. Methane
B. Water
C. Chlorine
D. Nitrogen

(v) An acid which has one replaceable hydrogen ions:

A. Formic acid
B. Sulphuric acid
C. Phosphoric acid
D. Phosphorus acid

(vi) The metal whose hydroxide is soluble in excess of sodium hydroxide is:

A. Lead
B. Magnesium
C. Copper
D. Calcium

(vii) The vapour density of benzene is 39, it's molecular mass is:

- **A.** **18.5**
- **B.** **78**
- **C.** **93**
- **D.** **80**

(viii) The gas which turns moist lead acetate paper silvery black is:

- **A.** **Ammonia**
- **B.** **Hydrogen sulphide**
- **C.** **Sulphur dioxide**
- **D.** **Hydrogen chloride**

(ix) The drying agent which forms an addition compound with ammonia gas is:

- **A.** **Calcium chloride**
- **B.** **Calcium oxide**
- **C.** **Phosphorus pentoxide**
- **D.** **Concentrated sulphuric acid**

(x) The pH of the solution which will liberate sulphur dioxide from sodium bisulphite is:

- **A.** **3**
- **B.** **8**
- **C.** **14**
- **D.** **12**

(xi) The gas released when copper turnings reacts with concentrated nitric acid is:

- **A.** **Nitric oxide**
- **B.** **Nitrogen dioxide**
- **C.** **Sulphur dioxide**
- **D.** **Sulphur trioxide**

(xii) The compound which can be used to obtain ethene from bromoethane is :

- **A.** **Alcoholic potassium hydroxide**
- **B.** **Calcium oxide**
- **C.** **Phosphorus pentoxide**
- **D.** **Aluminium oxide**

(xiii) The IUPAC name of iso butane is:

- **A.** **2,2-dimethylpropane**
- **B.** **2-methylpropane**
- **C.** **2-methylbutane**
- **D.** **3-methylbutane**

(xiv) The ion which will get discharged at the cathode during the elctrolysis of a solution containg the following ion is:

- **A.** **Copper**
- **B.** **Silver**
- **C.** **Sodium**
- **D.** **Potassium**

(xv) An alkaline earth metal found period 4 :

- **A.** **Calcium**
- **B.** **Magnesium**
- **C.** **Strontium**
- **D.** **Barium**

Question 2

(i) The following reactions are carried out: **[5]**

A: Nitrogen + Metal → Compound X

B: Compound X + warm water → Ammonia + Compund Y

C: Ammonia + Sulphuric acid → Ammonium sulphate

One metal that can be used for reaction A is magnesium.

(a) Write the formula of the compound X.

(b) Name another metal which can be used instead of magnesium in reaction A.

(c) Write a balanced chemical equation for reaction B.

(d) How is the ammonia gas collected?

(e) Which property of ammonia is demonstrated by reaction C?

(ii) Match the following Column A with Column B: **[5]**

Column A	Column B
(a) Zinc oxide	1. Nitrogen
(b) Copper hydroxide	2. Yellow when hot and white when cold
(c) Lead hydroxide	3. Pale blue
(d) Silver plating	4. Chalky white
(e) Triple covalent bond	5. Sodium argentocyanide

(iii) Complete the following by choosing the correct answer from the brackets: **[5]**

(a) Aqueous solution of sulphur dioxide gas is ____________ [neutral/ acidic]

(b) Ionic compounds are soluble in _________ [water/ organic solvents]

(c) The sulphide ore of lead is _________ [Cinnabar/ galena]

(d) The nitrate which on heating leaves a liquid residue ________ [ammonium/ mercury]

(e) Basic gas gives dense white fumes of _________ [ammonium chloride/ ammonium hydroxide] with hydrogen chloride.

(iv) Identify the following: **[5]**

(a) The gas produced when concentrated nitric acid reacts with sulphur.

(b) Hydrocarbons having carbon-carbon double or triple covalent bonds.

(c) The current needed for electroplating an article.

(d) The type of electrolyte which contains ions as well as molecules.

(e) A mixture which boils at a fixed temperature without any change in concentration or composition.

(v) (a) Draw the structural formula of the following:

(1) 2-methylbutanoic acid

(2) 2,2,2-trichloroethanal

(3) Propyne

(b) Write the IUPAC name of the following:

(1) The second member of alkene family with a bromine atom attached to the second carbon.

(2) Chloroform

SECTION B

(Attempt any four questions from this section)

Question 3

(i) Identify the anions present in each of the following compounds: [2]

(a) When dilute sulphuric acid is added to compound B a colourless gas with a suffocating odour is released which turns acidified potassium dichromate solution from orange to clear green.

(b) On addition of aqueous silver nitrate solution to a solution of compound D a curdy white precipitate is formed which dissolves in excess of ammonium hydroxide solution.

(ii) Write the products and balance the following equations: [2]

(a) $Cu + \text{conc } H_2SO_4 \rightarrow$

(b) $C + \text{conc } HNO_3 \rightarrow$

(iii) Arrange the following as per the instruction given in the brackets: [3]

(a) Na, Cl, Si, S (increasing order of oxidizing property)

(b) C, B, N, O (decreasing order of non-metallic character)

(c) Br, F, I, Cl (increasing order of electron affinity)

(iv) Fill in the blanks selecting the appropriate word from the given choice: [3]

(a) Pair of valence electrons which do not take part in bond formation __________ . (bond pair/lone pair)

(b) A molecule which has a double covalent bond ____________ (oxygen/ water)

(c) Ionic compounds conduct electricity in their molten state because they consist of free __________ (ions/ molecules)

Question 4

(i) For each of the substances given below, what is the role played in the extraction of Aluminium. [2]

(a) Gas carbon

(b) Cryolite

(ii) Calculate: [3]

(a) Give the empirical formula of N_2H_4

(b) A gas cylinder holds 5 g of hydrogen gas under certain conditions of temperature and pressure. If an identical cylinder with the same capacity can hold 85 g of gas 'P' under the same conditions of temperature and pressure, calculate the vapour density and molecular mass of gas 'P'.

(iii) The following questions are pertaining to the laboratory preparation of hydrogen chloride gas: [3]

(a) Write balanced chemical equation for the reaction taking place?

(b) Why is the gas not collected by downward displacement of water?

(c) How is the gas collected?

(iv) Explain the following: [2]

(a) Freshly prepared ferrous sulphate solution is used for brown ring test.

(b) Sulphuric acid cannot be obtained by dissolving sulphur trioxide in water.

Question 5

(i) (a) State the property of ammonia gas demonstrated in fountain experiment.

(b) A solution of hydrogen chloride gas in toluene is neutral. Why? [2]

(ii) Identify the cation present in the following compounds based on the observations:

(a) When ammonium hydroxide solution is added to a solution of compound X no precipitate is formed whereas when sodium hydroxide solution is added to the solution of compound X, a milky white precipitate is formed.

(b) A dull white precipitate is obtained when sodium hydroxide is added to an aqueous solution of 'Y'. [2]

(iii) Give balanced chemical equation for the following: [3]

(a) Sodium propionate is heated with soda lime.

(b) Ethyne is bubbled through bromine in carbon tetrachloride.

(c) Bromoethane is heated with hot concentrated alcoholic potassium hydroxide.

(iv) State one relevant observation for each of the following reactions: [3]

(a) Ammonia gas is passed over heated copper oxide.

(b) Molten lead bromide is electrolysed using graphite anode.

(c) Sodium hydroxide solution is added to aqueous zinc sulphate solution in small quantities.

Question 6

(i) Define: [2]

(a) Ore

(b) Matrix

(ii) Solve: [2]

200 cc of ethene is burnt in just sufficient air (containing 20% oxygen) to form carbon dioxide gas and steam according to the equation given:

$$2C_2H_4\ (g) + 3O_2\ (g) \rightarrow 2CO_2\ (g) + 2H_2O\ (l)$$

Calculate the volume of carbon dioxide formed and air needed.

(iii) Give reason for the following: [3]

(a) Electrolysis is an example of redox reaction.

(b) Electrolysis of acidified water is an example of a catalytic reaction..

(c) Soda lime is used for the preparation of methane from sodium acetate instead of sodium hydroxide.

(iv) Write balanced chemical equation for the following: [3]

(a) Dilute hydrochloric acid as a typical acid.

(b) Concentrated sulphuric acid as a dehydrating agent.

(c) Ammonia gas as a reducing agent.

Question 7

(i) 3.5 g of nitrogen combines with 2 g of oxygen to form an oxide of nitrogen. Determine the empirical formula of this oxide. [N=14 O=16] [3]

(ii) Identify the gas released in each of the following cases:

(a) Manganese dioxide is reacted with concentrated hydrochloric acid.

(b) Magnesium is reacted with cold dilute nitric acid. [2]

(iii) A blue crystalline solid M on heating produces a reddish brown gas N, a colourless, odourless gas O and leaves behind a black residue P.

(a) Identify M.

(b) Give one chemical test for gas N.

(c) Write balanced chemical equation for action of heat on M.

(iv) What would you observe when: **[3]**

(a) The ethane is burnt in excess of oxygen.

(b) Concentrated sulphuric acid is added to crystals of blue vitriol. **[2]**

Question 8

(i) Draw the electron dot and cross structure of the following molecules: **[2]**

(a) Methane

(b) Hydrogen chloride

(ii) Distinguish between the following pair of compounds: **[2]**

(a) Sodium hydroxide and ammonium hydroxide solution using copper nitrate solution.

(b) Ethane and ethyne using bromine water.

(iii) With respect to electrorefining of copper answer the following : **[3]**

(a) What change would you notice at the anode?

(b) Write equation for the reaction taking place at the cathode

(c) Write equation for the anode reaction.

(iv) Identify the functional group in the following organic compound: **[3]**

(a) C_2H_5COOH

(b) CH_3OH

(c) CH_3COCH_3